

CARVE

The Ultimate Parallax Occlusion Mapping (POM) Solution for Unity

Silhouette Parallax Occlusion Mapping for HDRP & URP · User Manual

Version 1.0.0 · Unity 6.0 LTS or newer · HDRP and URP

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1. Overview & Features

Carve gives flat surfaces real depth. It is parallax occlusion mapping (POM) built to hold up everywhere a classic POM shader falls apart: the edges of a mesh are carved into the shape of the relief, props meet the relief at its true depth, and light, shadows and reflections follow every crack. It works on modular kits, curved props and meshes with poor or missing UVs, and a scene-wide Quality Manager keeps it inside your frame budget.



The Showcase scene with Carve on: relief with real depth, carved edges and self-shadowing on plain box and column meshes.

Real depth, real outlines

- **Carved silhouettes:** box and modular edges show the relief's outline instead of a straight line; rocks, columns and pipes get uneven outlines.
- **Real depth:** props, characters and decals meet the carved surface at the right depth, and SSAO, fog and contact shadows see it too.
- **Depth in centimeters:** 5 cm of relief is 5 cm on any mesh, scale or tiling.
- **Anti-tiling and detail relief** for big surfaces, **height-blended layers** you paint like terrain, **Carve Decals**, and one-click **relief colliders and meshes**.

Light that understands the relief

- Soft self-shadows from the sun, relief occlusion, no glowing cracks in reflections, cliff shading on steep walls, and cast shadows with the carved shape.

Fast by design

- Auto LOD and Pyramid tracing spend steps only where they show. The Carve Quality Manager sets every material in a scene with one preset and can scale Carve's work in Play mode to hold a frame rate. A built-in cost meter shows what Carve costs in your scene.

Works on anything

- Triplanar mapping for meshes without good UVs, one-click silhouette data for modular kits, no mesh tangents needed, and a Carve POM node for your own Shader Graphs.

No setup headaches

- 13 ready-to-use materials, a height map generator, a one-click converter from Lit (HDRP and URP), health checks with one-click fixes, and the Carve Hub window for everything.

2. Requirements

Item	Requirement
Unity	Unity 6.0 LTS (6000.0) or newer.
Render pipeline	High Definition Render Pipeline 17 or newer, or Universal Render Pipeline 17 or newer (the versions that ship with your Unity 6 editor). One package holds both; the shaders for the pipeline in use are picked up automatically.
URP rendering paths	Forward, Forward+ and Deferred.
Shader Graph	Installed with HDRP and URP. Needed for Carve Decals, the Carve POM node and its example graphs.
Anti-aliasing	TAA is recommended on cameras: relief detail is finer than a pixel. The Setup Wizard sets it for you.
Input	The demo scenes work with the legacy Input Manager or the Input System package.

Carve has no third-party dependencies. Every texture, mesh and material in the package was made procedurally for Carve.

HDRP and URP

One package for both pipelines. The HDRP shaders, materials and demo scenes are in the package folders; the URP ones come in `Carve URP Content.unitypackage` (in the package root), which the Setup Wizard imports in a URP project with one click (or **Tools > Carve > Import URP Content**). An HDRP project never sees URP files, and the editor code compiles only for the pipeline you have.

- The URP shaders write real depth in every pass (color, depth prepass, depth normals, shadow maps), so props, SSAO, decals and shadows see the carved surface, as in HDRP.
- Self-shadowing, relief occlusion, specular occlusion, cliff shading, triplanar, anti-tiling, detail relief, layers, Auto LOD, pyramid tracing and the Quality Manager work in both pipelines.
- URP has one occlusion term for ambient light and reflections, so Carve folds its reflection trace (no glowing cracks) into it.
- TAA: HDRP's camera TAA, or URP's Temporal Anti-aliasing on the camera (Unity 6).

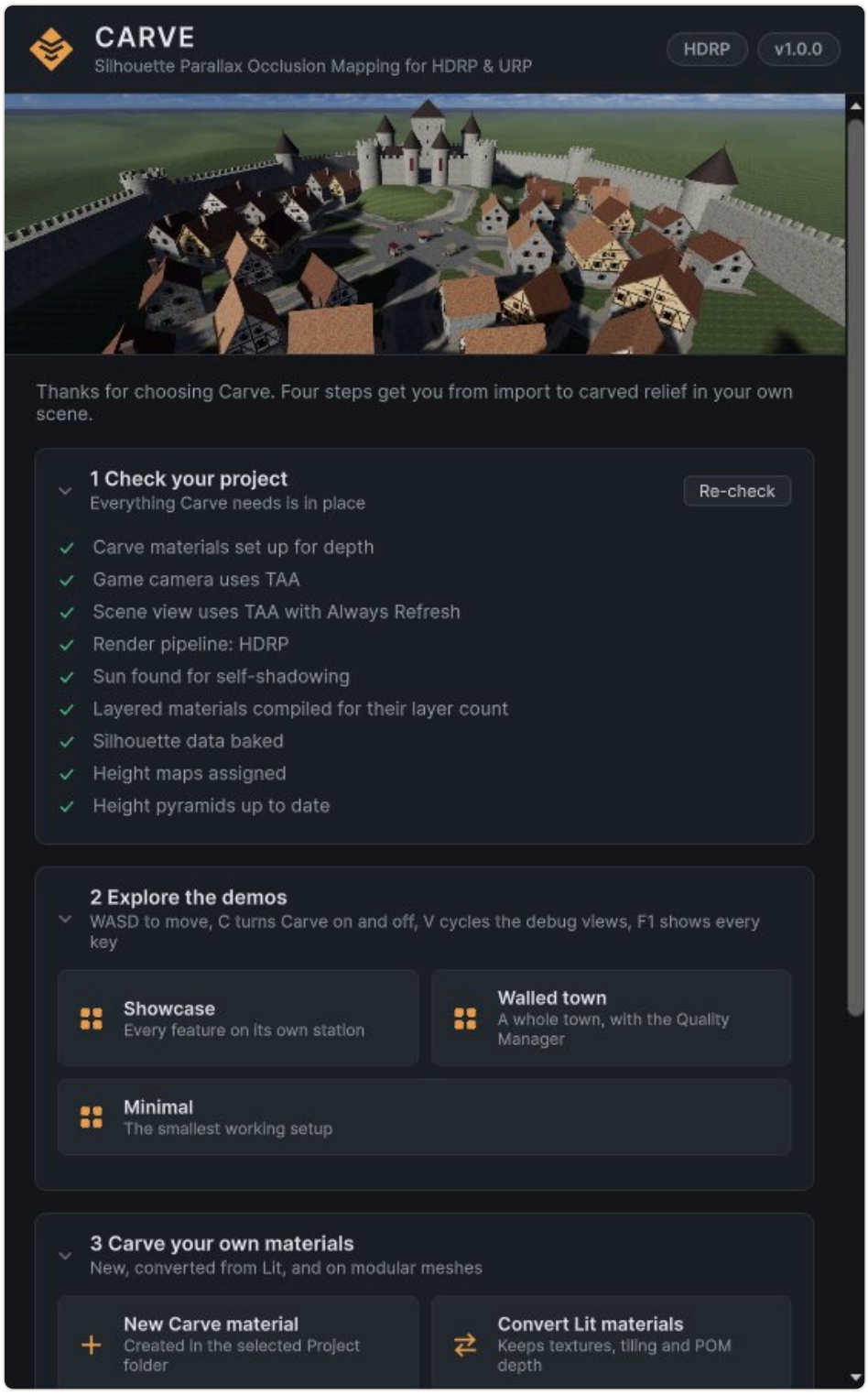
3. Quick Start

1. Import the package. The **Setup Wizard** opens: fix anything it lists with its buttons. In URP the first fix is **Import Carve's URP content**: it imports `Carve URP Content.unitypackage` (URP shaders, materials and demo scenes).
2. Open a demo scene (`Carve_Showcase` or `Carve_Village`; in URP the `_URP` versions) and press Play to see every feature. `C` switches Carve on and off.
3. In your own project, create a material with **Carve/HDRP/Carve Lit** or **Carve/URP/Carve Lit** (or select Lit materials and use **Assets > Carve > Convert to Carve**).
4. Assign a **Height Map** (linear grayscale, white = top) and set **Depth (cm)**. No height map? Click **Generate Height Map** to make one from the normal map.
5. On boxes and modular meshes set **Edge Silhouettes = Baked** and bake the meshes (Carve Hub > Meshes). On rocks and columns set **Curved Silhouettes = Auto**.
6. Add a Quality Manager to the scene (**GameObject > Carve > Quality Manager**) and pick a preset.

> The fastest way to try Carve on an existing scene: open **Tools > Carve > Carve Hub**, go to **Convert**, convert your Lit materials as copies, and compare.

4. Setup Wizard

The Setup Wizard opens once per project, the first time Carve is imported. Reopen it any time from **Tools > Carve > Setup Wizard**.



The Setup Wizard: project checks, demo scenes, first steps and support.

1. **Check your project** runs the same health checks as the Carve Hub on your project and the open scene, each with a fix button: render pipeline, the URP content (URP only), a sun for self-shadowing, game camera TAA (and Scene view TAA in HDRP), HDRP material surface options, URP's Decal renderer feature when Carve Decals are used, baked silhouette data, missing height maps and height pyramids.
2. **Explore the demos** opens the demo scenes.
3. **Carve your own materials**: a new Carve material, the converter, silhouette baking and the Quality Manager.
4. **Learn more**: the Carve Hub, this manual and support.

5. Carve Lit Materials

Carve/HDRP/Carve Lit and **Carve/URP/Carve Lit** are full lit surfaces with Carve's relief built in. Both have the same settings and property names, so materials, presets and scripts work the same way in either pipeline. The HDRP inspector keeps HDRP's surface options; the URP inspector adds Render Face, render queue and GPU instancing. Every setting has a tooltip.

Surface

Setting	What it does
Base Map	Albedo (RGB). Its tiling and offset apply to every map.
Normal Map	Tangent-space normal map and its strength.
Mask Map	Packed mask (HDRP layout, in both pipelines): R metallic, G ambient occlusion, A smoothness.
Metallic, Smoothness	Multiply the mask map's channels.
Mapping	UV, Triplanar Object or Triplanar World (see section 7).

Relief

Setting	What it does
Height Map	Linear grayscale: white = top, black = deepest. 16-bit (R16) gives the smoothest relief. Empty with a normal map assigned shows **Generate Height Map**.
Depth (cm)	Real depth of the relief in centimeters, independent of UV scale and tiling.
Steps (min / max)	Ray-march steps: min looking straight at the surface, max at grazing angles.
Refine Steps	Extra steps that sharpen the hit point.
Tracing	Linear (fixed steps) or Pyramid (marches only between the lowest and highest relief under each pixel: fewer steps, never coarser). The pyramid bakes automatically.
Auto LOD	Caps the steps at this many per screen pixel the ray crosses. Distant and shallow relief costs a few steps; close-ups are unchanged. 0 = off.
Distance Fade (m)	The relief fades out between these distances.
Stability	Less shimmer without TAA (Scene view, VR, MSAA): a slightly coarser height and softer tiny shadows. 0 = sharpest.
Anti-Tiling, Cell Size	See section 8.

Silhouettes

Setting	What it does
Edge Silhouettes	Off, UV Tile (every 0-1 UV tile is a face: Unity's cube, many modular kits) or Baked (face outlines baked into the mesh).
Curved Silhouettes	Off, Auto (measured on the fly, no setup) or Baked (per-vertex curvature from the mesh baker, smoothest).
Curvature Scale	Exaggerates or softens curved outlines.

Shadows & Occlusion

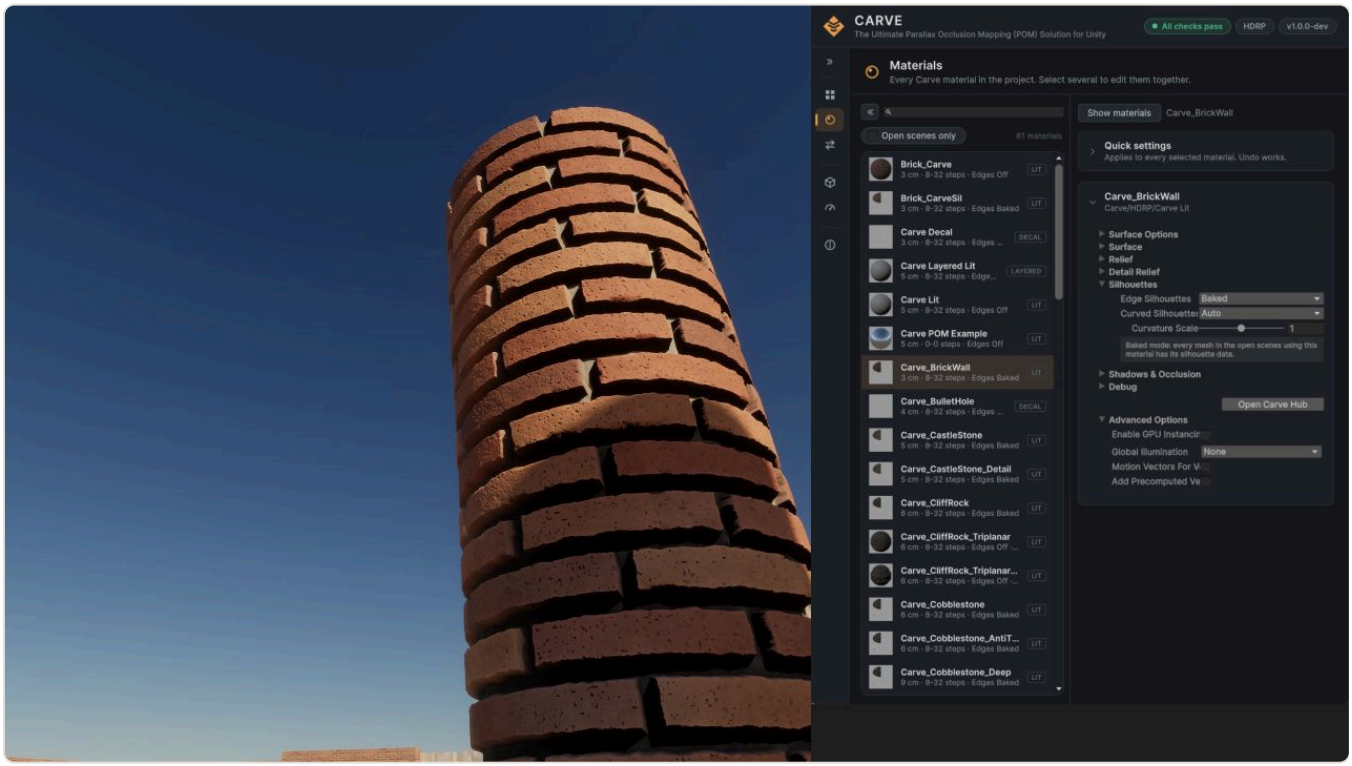
Setting	What it does
Self Shadow, Shadow Softness (°), Shadow Steps	Soft shadows of the relief onto itself from the sun. Softness is the light's angular size.
Relief Occlusion, Occlusion Radius (cm)	Darkens deep, enclosed gaps in ambient light.
Specular Occlusion	Blocks reflections that would leak out of cracks.
Cliff Shading, Cliff Threshold	Shades steep relief walls like real geometry and hides texture stretching.
Cast Shadow Steps	Share of the view steps used for shadow maps (0.5 is usually enough).

Debug and Detail Relief

Debug View shows one part of Carve on this material (section 18). **Detail Relief** is described in section 8.

> HDRP only: Carve needs HDRP's **Depth Offset** on and cuts outlines itself, so HDRP **Alpha Clipping** stays off. These are set automatically when a material switches to Carve; if something turns them back, the inspector and the health checks show a **Fix** button.

6. Carved Silhouettes & Modular Meshes

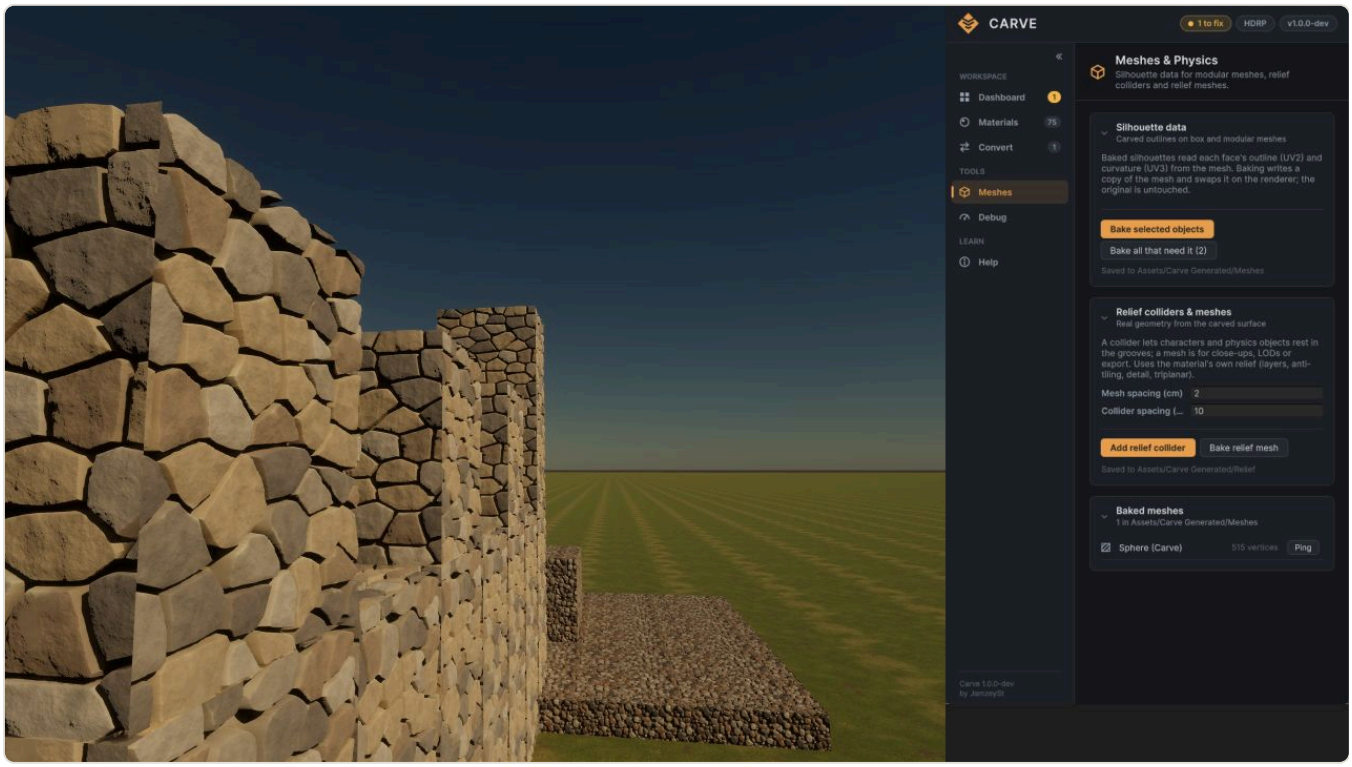


Curved Silhouettes (Auto) on a column: the outline follows the bricks.

A classic POM shader stops at the mesh outline, so every edge is a straight line. Carve cuts rays that leave a face, which carves the relief's shape into the outline.

- **UV Tile:** each 0-1 UV tile counts as a face. Works with Unity's cube and kits whose faces each fill the UV square.

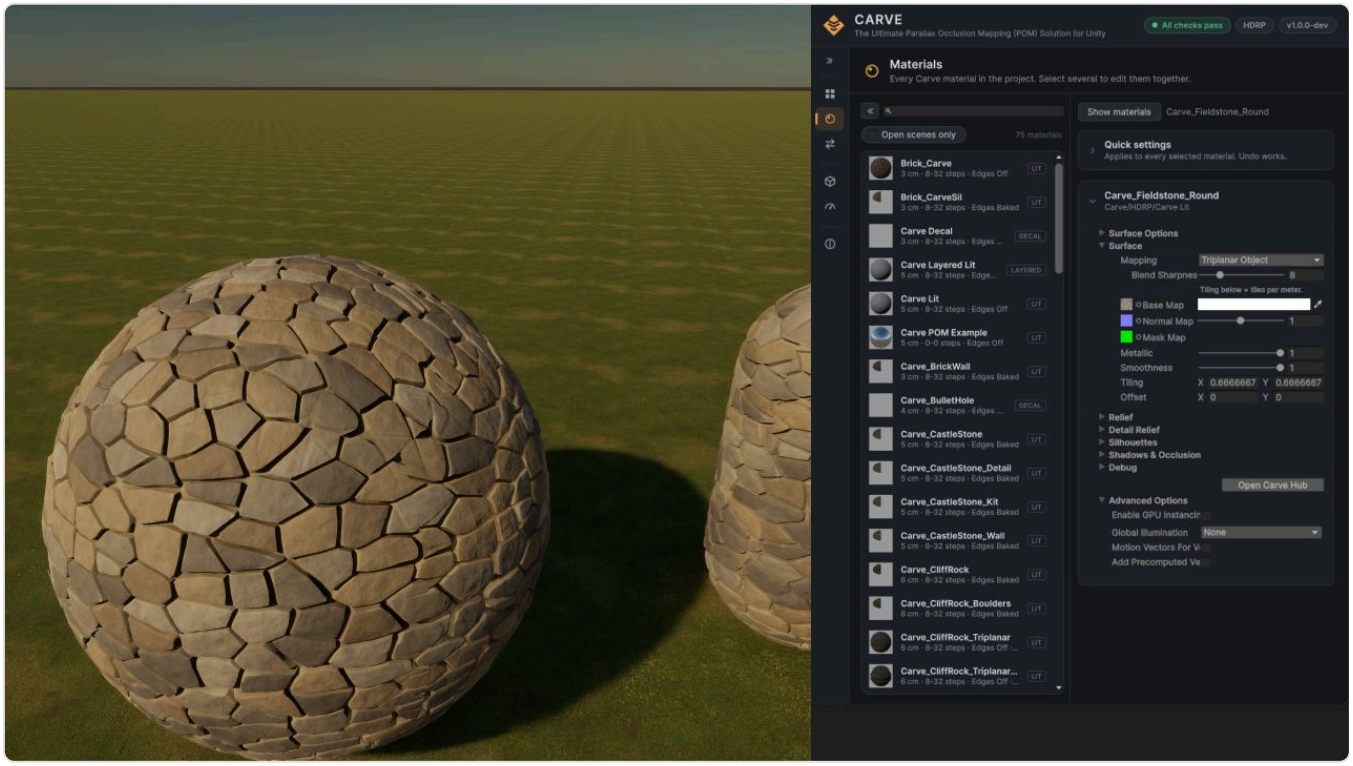
- **Baked:** for any mesh. Select objects and use **Tools > Carve > Bake Silhouette Data (Selected)** or the Carve Hub's **Meshes** page. Carve writes a copy of each mesh with the face outlines (UV2) and curvature (UV3) to `Assets/Carve Generated/Meshes` and swaps it in. Your original mesh is not changed.
- **Curved Silhouettes:** Auto needs no setup; Baked uses the curvature from the same bake.



A wall kit of plain 1 m cubes after baking: every block edge is carved.

The health checks warn when a material uses Baked silhouettes on meshes that were never baked, with a **Bake now** button.

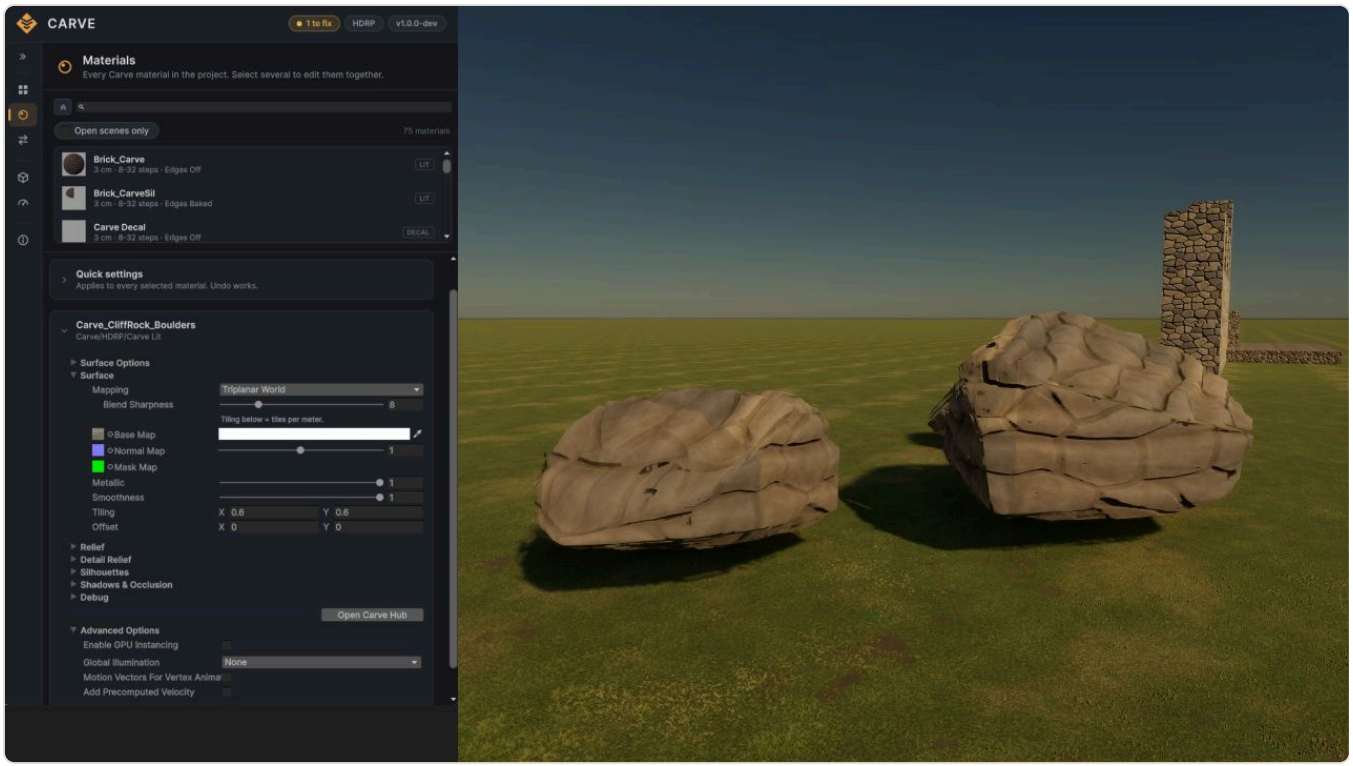
7. Mapping: UV and Triplanar



Unity's sphere with Triplanar Object mapping: no pole swirl, no seam.

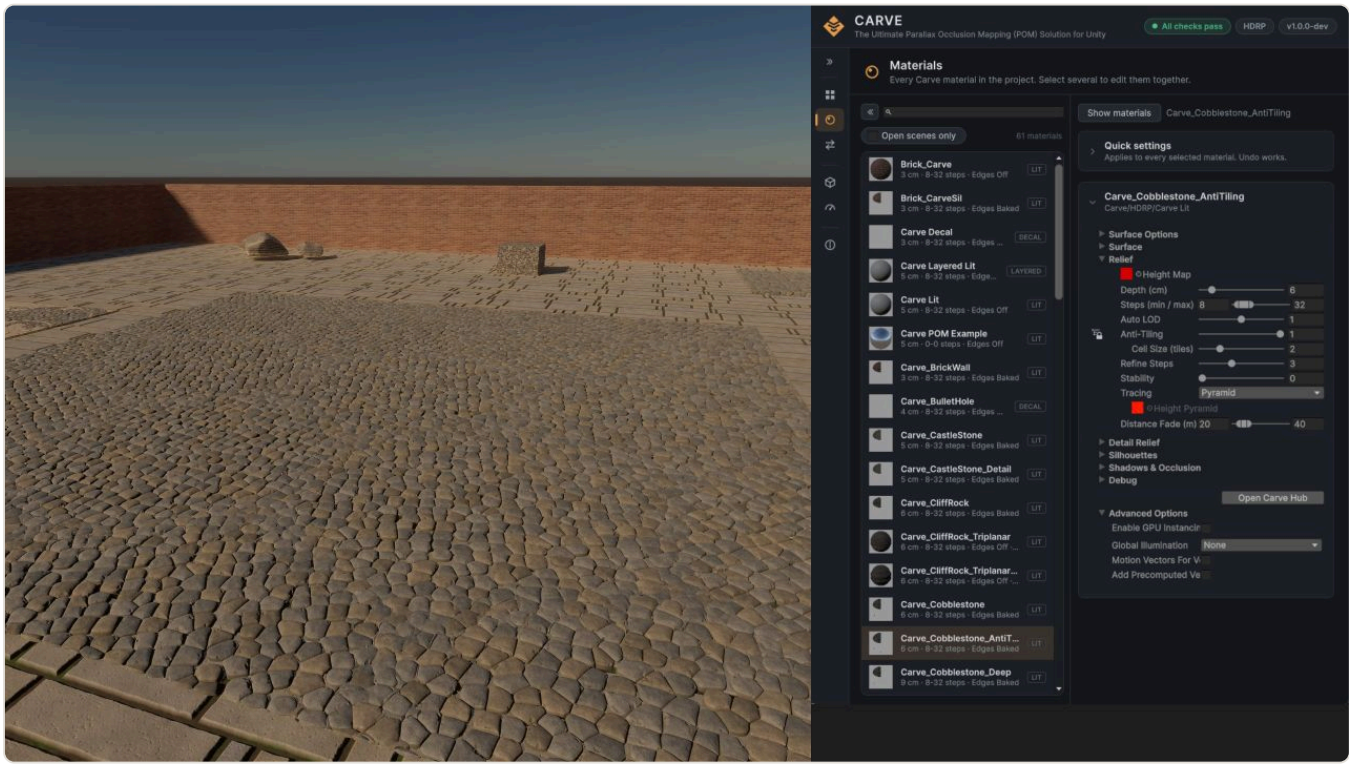
- **UV:** the mesh's own UVs. Tiling is per UV tile.
- **Triplanar Object:** projected from three sides in object space and moves with the object. Use it on rocks, spheres and scans with bad or no UVs. Tiling is in tiles per meter.
- **Triplanar World:** the same in world space, so textures continue across neighbouring objects.
- **Blend Sharpness** sets how quickly the projections hand over.

In triplanar modes curved outlines are still carved; edge silhouettes need UV faces.



Rocks with spherical UVs switched to Triplanar World.

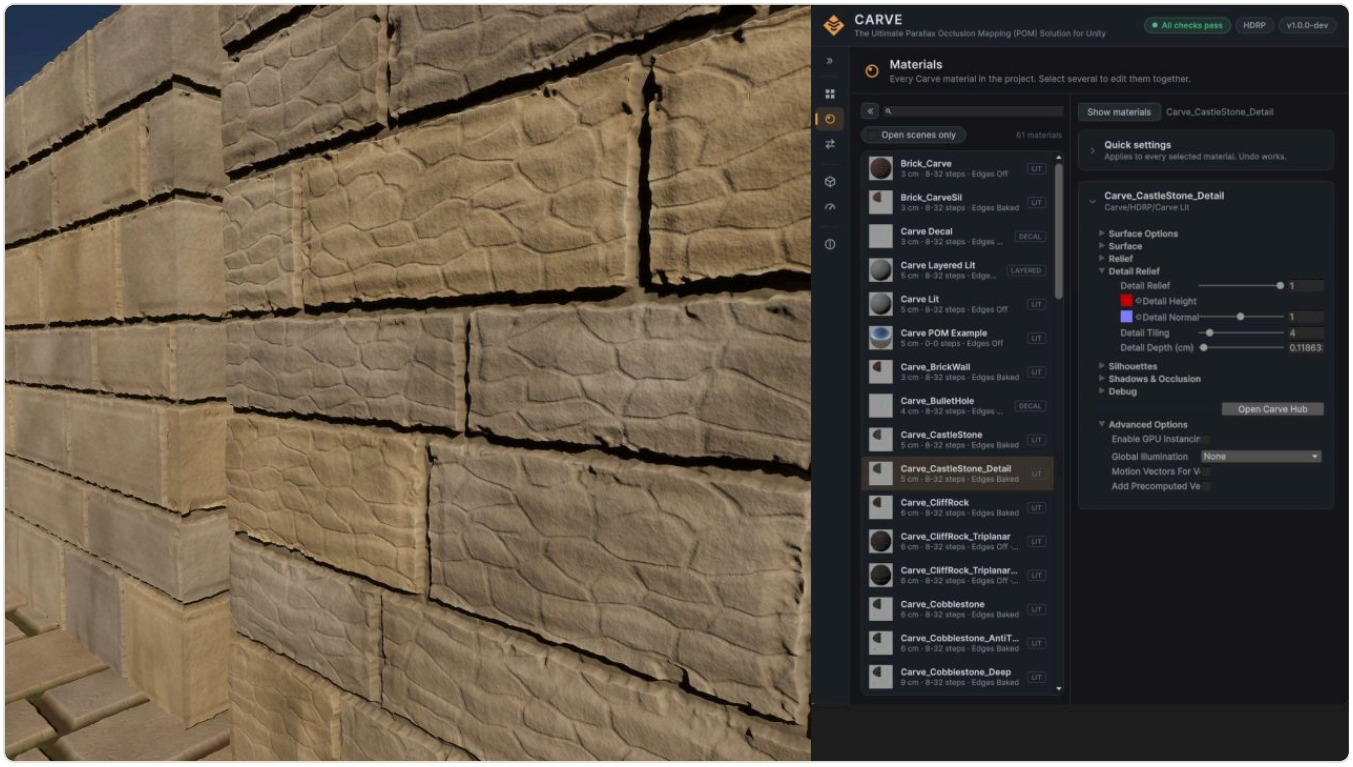
8. Anti-Tiling & Detail Relief



Anti-tiling on a large ground: no repeating pattern, and stones stay whole.

Anti-Tiling breaks up repetition on large floors and walls. Each hex cell gets its own offset and rotation of the texture, and the relief follows, so nothing floats or mismatches. Where cells meet, the seam runs around whole stones. **Cell Size** is in texture tiles: larger cells mean fewer seams and less variation. It costs three height samples per step, so use it where repetition shows.

Detail Relief adds a second, finer relief (pores, cracks, grit) on top with its own **Detail Height**, **Detail Normal**, **Detail Tiling** and **Detail Depth (cm)**. Unlike a detail normal map it has real depth: it self-shadows and catches light. **Generate from Detail Normal** makes its height map for you.



Detail Relief on castle stone: fine cracks with their own shadows.

9. Carve Layered Lit & Layer Painting



Height-blended layers: pebbles settle into the gaps of cracked mud.

Carve Layered Lit (`Carve/HDRP/...` or `Carve/URP/...`) blends 2 to 4 surfaces on one mesh. Where they meet, the taller relief wins, so sand settles between stones and rocks poke through moss. The blend has real depth: it casts shadows and gets carved silhouettes.

Setting	What it does
Layers	2, 3 or 4. Layer 0 is the base; layers 1-3 are painted over it.
Mask Source	Mask Map (texture, R/G/B = layers 1/2/3), Vertex Color (Polybrush or any vertex painter) or Both.
Mask UV	UV0, or UV1 (lightmap UVs) for meshes whose UV0 tiles.
Height Blend	1 = the taller surface wins; 0 = plain cross-fade.
Blend Softness	Width of the transition, in relief height.
Per layer	Maps, tiling, **Height Scale** (share of Depth) and **Height Offset**. A raised flat layer (no height map) fills the gaps of the layer below: sand, water, snow.

Painting layers

Select objects with a Carve Layered Lit material and click **Paint Layers in Scene View**. Brush like terrain painting: `1`-`4` pick a layer, `Shift` erases, `[` `]` change the size. Soft brushes let the relief decide the boundary; hard ones give crisp borders. Undo, smooth, fill and pen pressure are supported. The mask is a texture in `Assets/Carve Generated/Masks`, updated live. For shared materials Carve offers a unique copy per object.

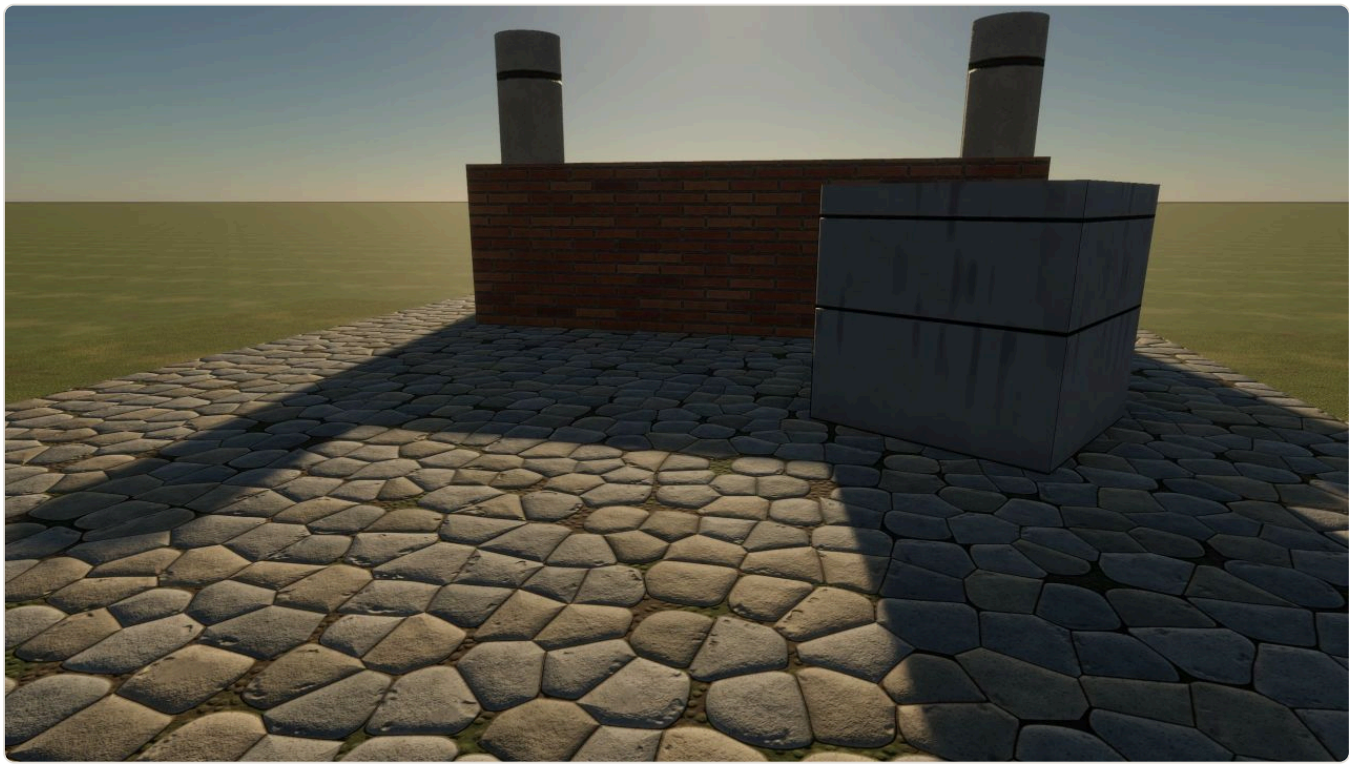
> Carve compiles each layered material for its layer count, so 2 layers cost much less than 4. Change the count from scripts with `CarveLayers.SetLayerCount` so this stays in sync.

10. Shadows, Occlusion & Light



Self Shadow: stones shade each other, with soft edges.

- **Self Shadow** marches toward the sun. The penumbra widens with distance like a real shadow; **Shadow Softness** is the sun's angular size.
- **Relief Occlusion** darkens cracks and gaps in ambient light, and bends ambient light away from walls.
- **Specular Occlusion** stops wet or shiny relief from reflecting the sky out of gaps (no glowing cracks).
- **Cliff Shading** shades steep relief walls like real geometry instead of showing stretched texture.
- **Cast shadows** have the carved outline, and other objects' shadows bend over the relief. **Cast Shadow Steps** sets the cost.



Shadows carry the carved shape, and fall over the relief.

The sun is `RenderSettings.sun`, else the brightest active directional light. Set `CarveSun.Override` to choose one yourself.

11. Carve Decals



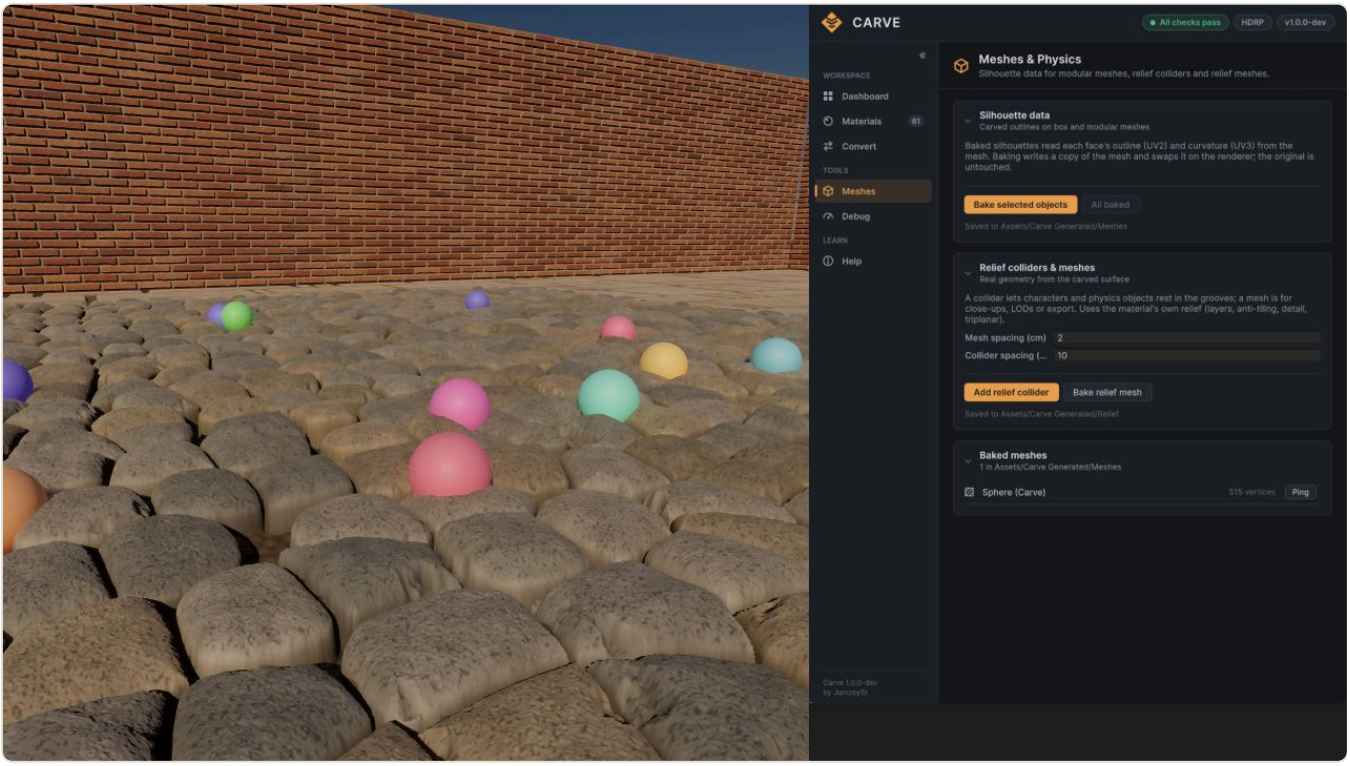
Carve Decals on a brick wall: holes and cracks that look cut in.

Carve Decal (`Carve/HDRP/Carve Decal`, `Carve/URP/Carve Decal`) makes cracks, bullet holes, grates and manhole covers look cut into any surface, Carve or not, with lit walls, dark depths and sun shadows inside. Four decals are included for each pipeline (`Materials/HDRP/Decals`, `Materials/URP/Decals`). Use them with the pipeline's Decal Projector. In URP, add **Renderer Feature > Decal** to your Universal Renderer (the health checks remind you).

Setting	What it does
Depth (m)	How deep the cut goes.
Surface Coverage	0 = only the holes show (the surface below stays visible); 1 = the decal's own top is painted too (plates, covers).
Rim Softness, Cavity Darkening	Edge blend and darkening of the depths.
Steps, Auto LOD, Fade, Self Shadow	As on Carve Lit.

> Decals give parallax depth but can't write depth (an engine limit for projected decals), so props don't sink into them.

12. Relief Colliders & Relief Meshes



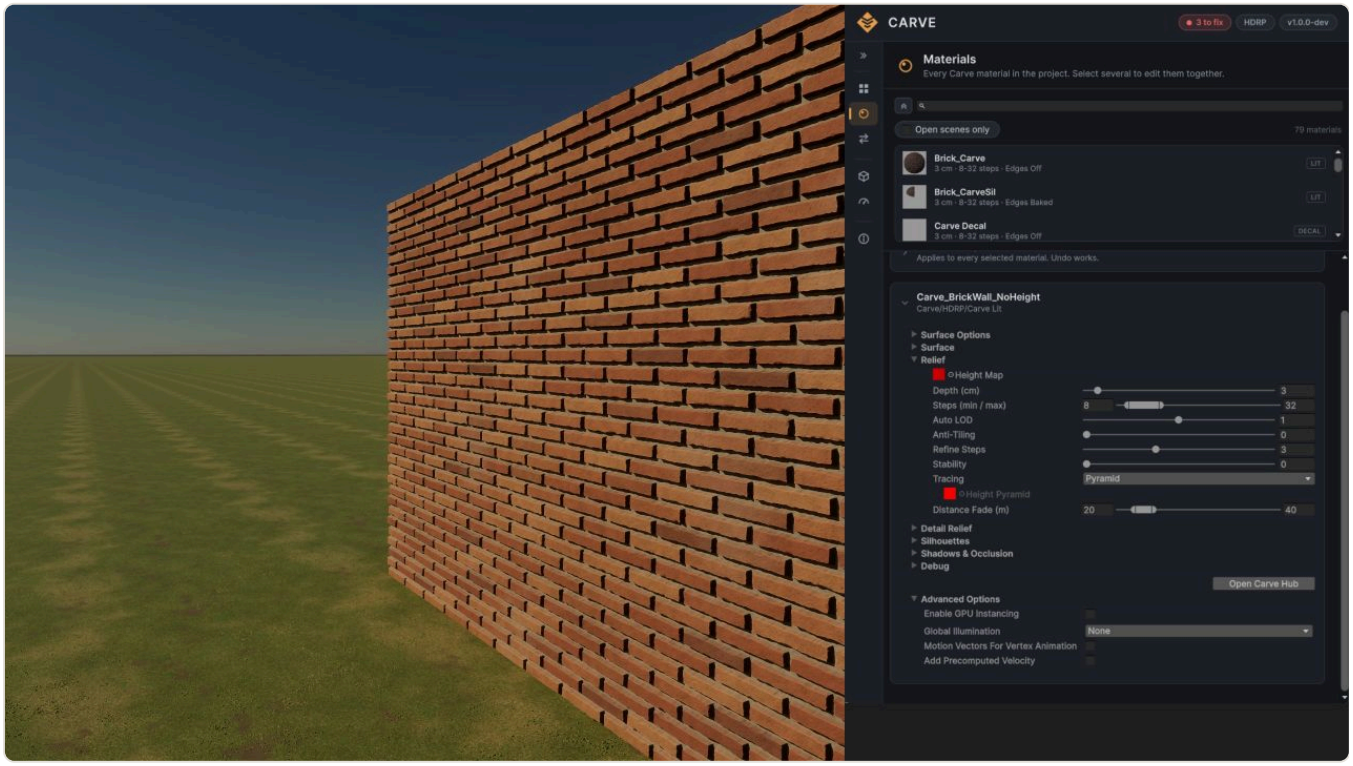
A relief collider: physics balls settle in the grooves.

Turn the carved surface into real geometry with one click (**GameObject > Carve > Add Relief Collider** / **Bake Relief Mesh**, or the Hub's Meshes page):

- **Relief Collider:** a MeshCollider that follows the relief, so props and characters rest in the grooves.
- **Relief Mesh:** a detailed mesh for close-ups, LODs or export.

The height comes from the material's own shader code (layers, anti-tiling, detail, triplanar), so the geometry matches what you see. Spacing is set on the Meshes page. Output goes to `Assets/Carve Generated/Relief`.

13. Height Maps & Height Pyramids



A brick material with no height map: one click generates it from the normal map.

Generate Height Map appears when Base Map and Normal Map are set and Height Map is empty. It integrates the normal map's slopes into a height map that tiles like the source, saves it as a 16-bit PNG next to the albedo and assigns it. The ▾ button opens its options:

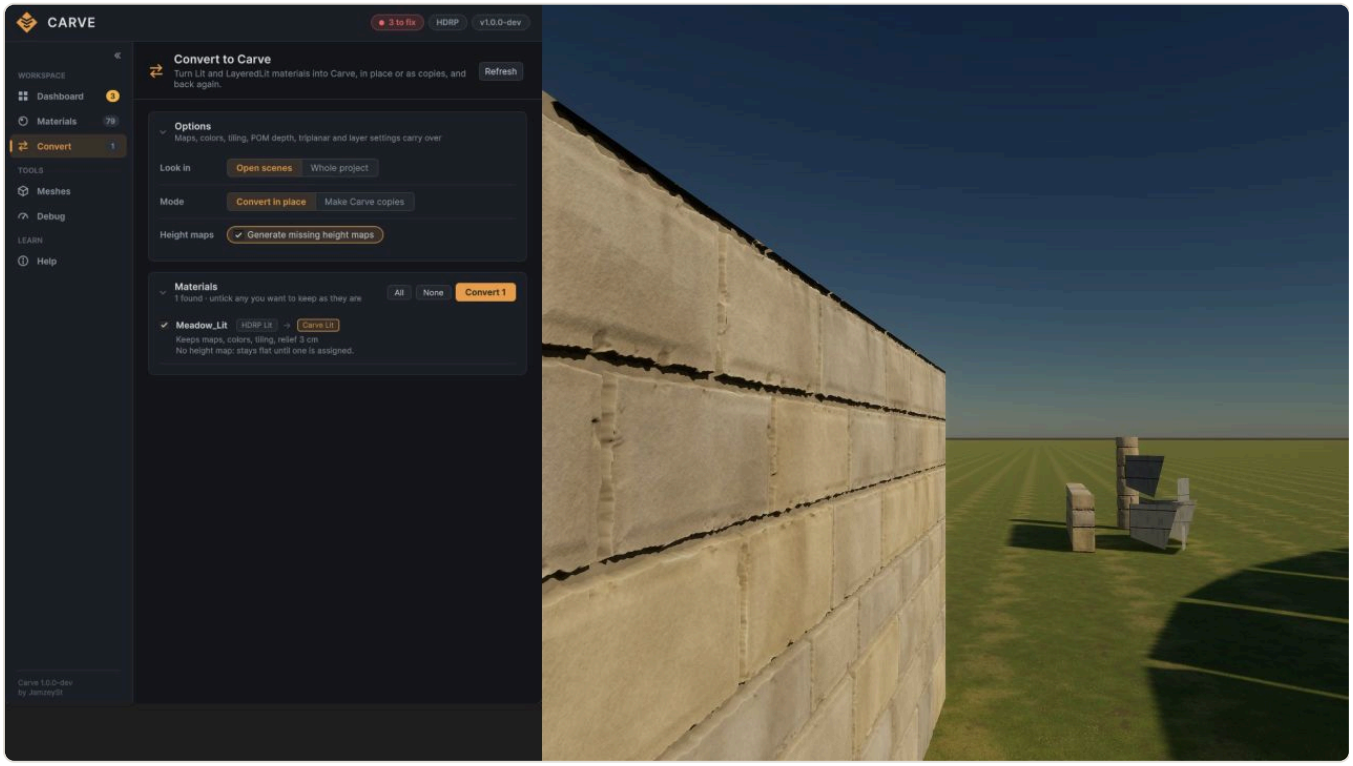
- **Smoothing** removes the normal map's fine grain, which would become tiny pits.
- **Albedo detail** adds fine detail from the albedo (usually best at 0).
- **Normal map green**: Auto detects DirectX (Y-) normal maps; Unity uses OpenGL (Y+).
- **Invert** swaps high and low.

Generated heights are a good start, not a scan: hand-made or scanned height maps are still best when you have them.

Height pyramids

Pyramid tracing uses a small min/max pyramid of the height map (about 2.8 MB for a 2K map). It bakes automatically when a height map is assigned and rebakes when the height map changes. Bake by hand with **Assets > Carve > Bake Height Pyramid** (for the Carve POM node) or **Tools > Carve > Bake Height Pyramids (Selected Materials)**. Pyramids live in `Assets/Carve Generated/Pyramids`.

14. Converting Lit Materials



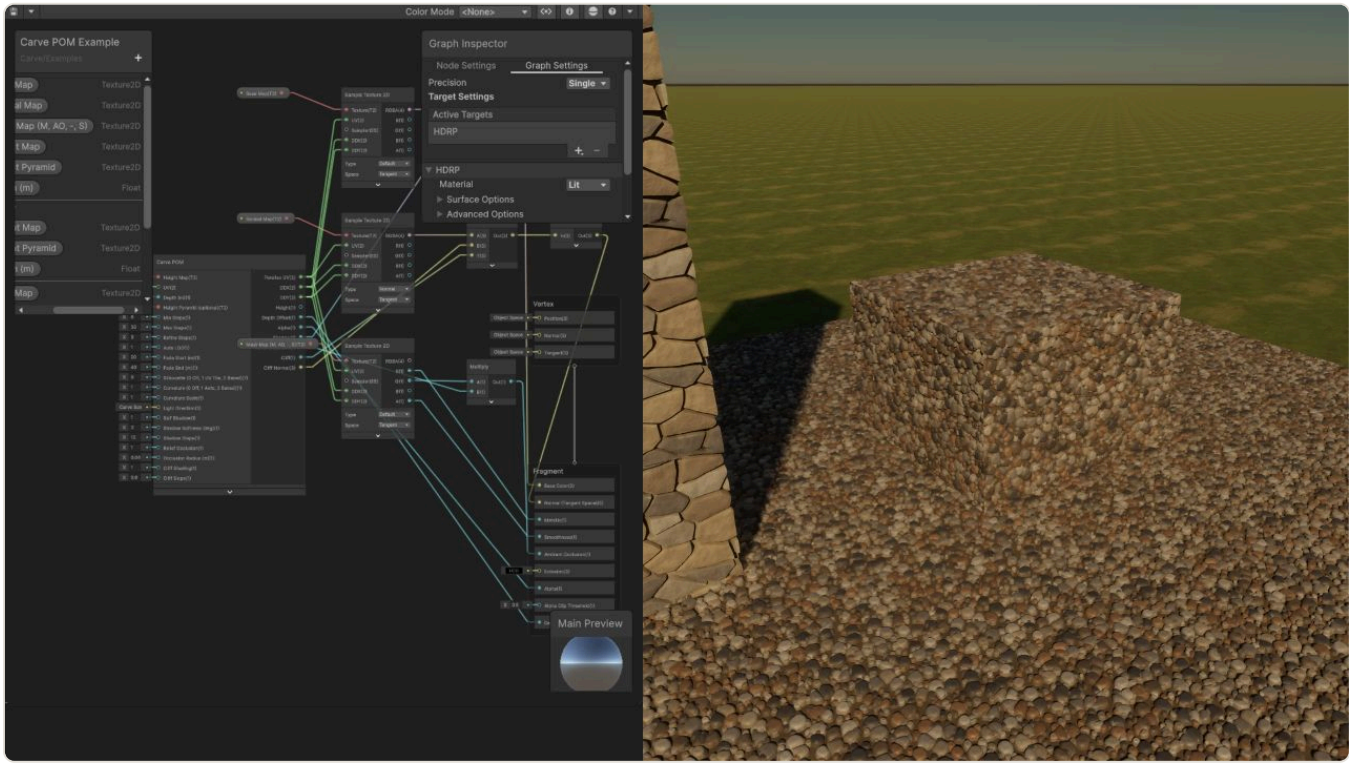
The Convert page: an HDRP Lit wall with Unity's POM, converted to Carve.

Convert HDRP **Lit** and **LayeredLit**, or URP **Lit**, materials to Carve from the Carve Hub's **Convert** page, or select materials and use **Assets > Carve > Convert to Carve**.

- **In place** (Undo works) or **as copies**, optionally swapped into the open scenes.
- Maps, colors, tiling, smoothness and metallic remaps, triplanar settings, layer heights and masks carry over.
- Unity's POM depth depends on the mesh's UV size and scale; Carve measures them on the scene objects using the material and converts it to real-world depth, so surfaces keep their look.
- Anything that can't carry over is listed before you convert. Materials without a height map can get one generated.
- URP Lit: its metallic/smoothness and occlusion maps are packed into one Carve mask map (a new texture next to the material), and its parallax height and amount become a real-world depth.
- **Convert back to Lit** returns a material to the pipeline's Lit.

> Converted materials can look darker than Lit, because Carve adds self-shadowing and relief occlusion. Turn those down to compare like for like.

15. Carve in Your Own Shader Graphs



The example graph built with the Carve POM node, next to the floor it drives.

The **Carve POM** node (**Create Node > Carve > Carve POM**) gives your own graphs the same relief as Carve Lit: depth in meters, carved outlines, sun self-shadow, relief occlusion and cliff shading. Open the example graph from the Hub's Help page.

1. Add the node and connect your Height Map (UV defaults to UV0).
2. Sample your textures at **Parallax UV** with Mip Sampling Mode = **Gradient**, wired to **DDX** and **DDY** (no mip seams).
3. Multiply Base Color by **Shadow** and Ambient Occlusion by **Relief Occlusion**.
4. HDRP: enable **Depth Offset** and **Alpha Clipping** in Graph Settings, then connect **Depth Offset** and **Alpha**.
5. URP: enable **Alpha Clipping** and connect **Alpha**. URP Shader Graph has no depth offset, so a URP graph gets the relief, outlines and lighting but not real depth (Carve Lit in URP does write real depth). See the URP example graph.
6. Faster: bake a height pyramid (right-click the height map > Carve > Bake Height Pyramid) and plug it into **Height Pyramid**.

Inputs also include Depth, steps, Auto LOD, fade, Silhouette (UV Tile / Baked), Curvature (Auto / Baked), Light Direction (defaults to the Carve sun), and shadow, occlusion and cliff settings. Outputs: Parallax UV, DDX, DDY, Height, Depth Offset, Alpha, Shadow, Relief Occlusion, Cliff, Cliff Normal. The node covers single-texture relief; triplanar, anti-tiling, detail relief and layers are Carve Lit / Carve Layered Lit features.

16. Quality Manager & Adaptive Performance



The Quality Manager inspector with a Custom setup and its Live readout.

Add one per scene with **GameObject > Carve > Quality Manager**. It sets how much work every Carve material in the scene does.

Presets

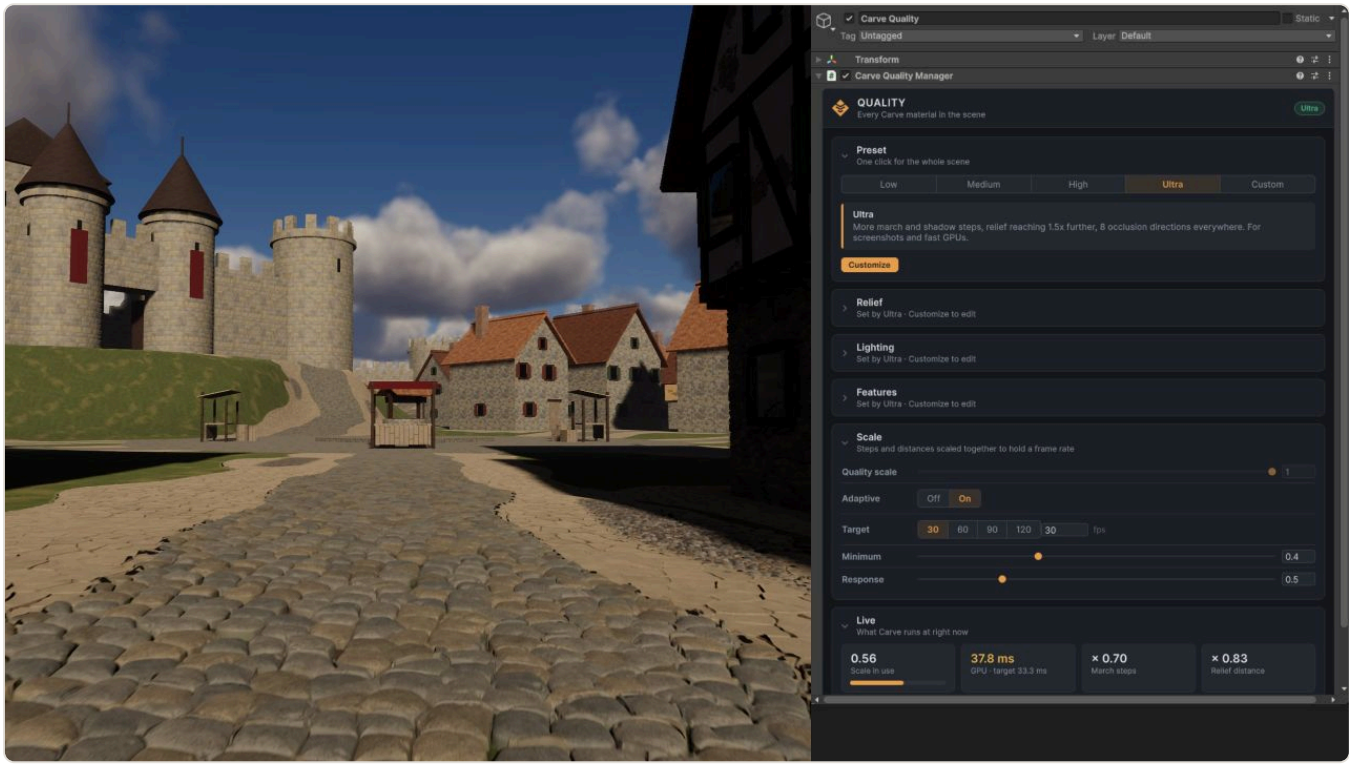
Preset	What it does
Low	Fewer steps, relief to half distance, shading extras within 20 m, no relief occlusion or cliff shading. For slower GPUs and huge scenes.
Medium	Slightly fewer steps, relief to 3/4 distance, shading extras within 40 m. Every feature stays on.
High	Every material exactly as authored. The default.
Ultra	More march and shadow steps, relief reaching 1.5x further, 8 occlusion directions everywhere. For screenshots and fast GPUs.
Custom	Your own settings. **Customize** copies the current preset into Custom.

The settings: **March steps** and **Relief distance** (scales every material's fade distances), **Shadow steps**, **Extras distance** (self-shadow, occlusion and cliff shading end at that many meters), **Occlusion directions** (Auto / 4 / 8), and scene-wide strength of **Self-shadow**, **Relief occlusion**, **Cliff & specular** and **Anti-tiling**. All are relative to each material's own settings.

Scale and adaptive

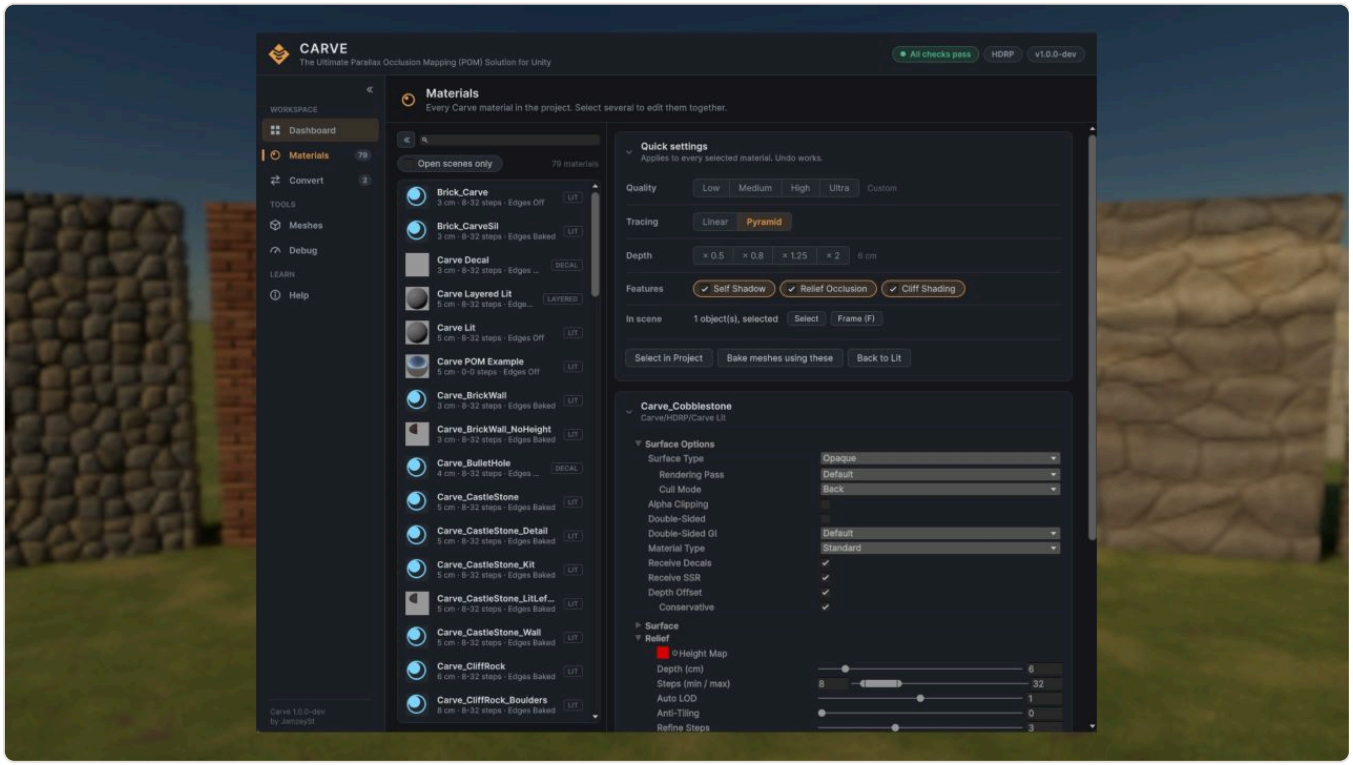
Quality Scale scales march steps, shadow steps, relief distance and extras distance together. With **Adaptive** on, in Play mode Carve lowers the scale when frames run over the **Target Frame Rate** and raises it again when there is room; it keeps the frame between 90% and 100% of the budget, lowering fast and raising three times slower so it settles. **Minimum Scale** is the floor; **Response Speed** sets how fast it moves. Your resolution never changes.

The **Live** card shows the scale in use, the measured frame time (GPU time where the platform reports it), and the steps and distance in effect. In the demo scenes `F2` cycles presets and `F3` switches adaptive.



Adaptive at work in the walled town: the scale drops in heavy views and recovers.

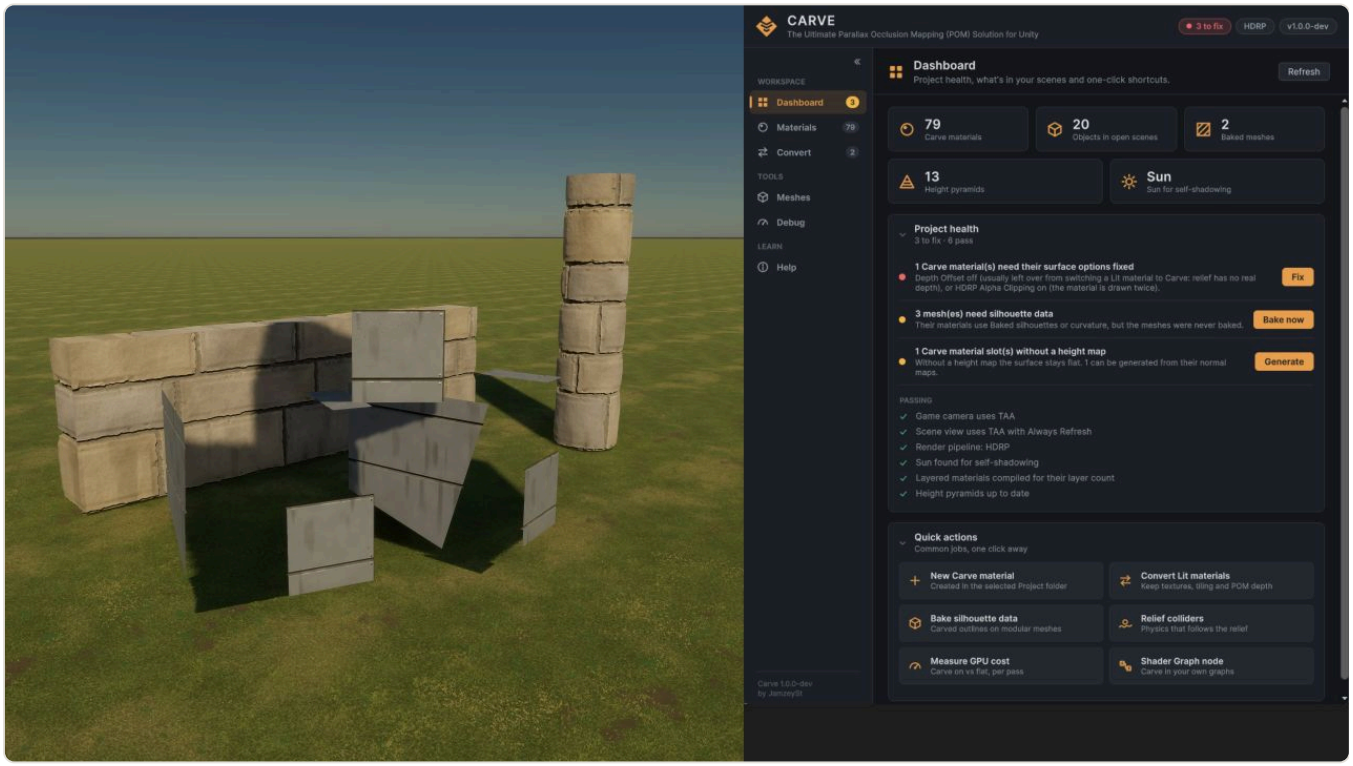
17. The Carve Hub



The Carve Hub.

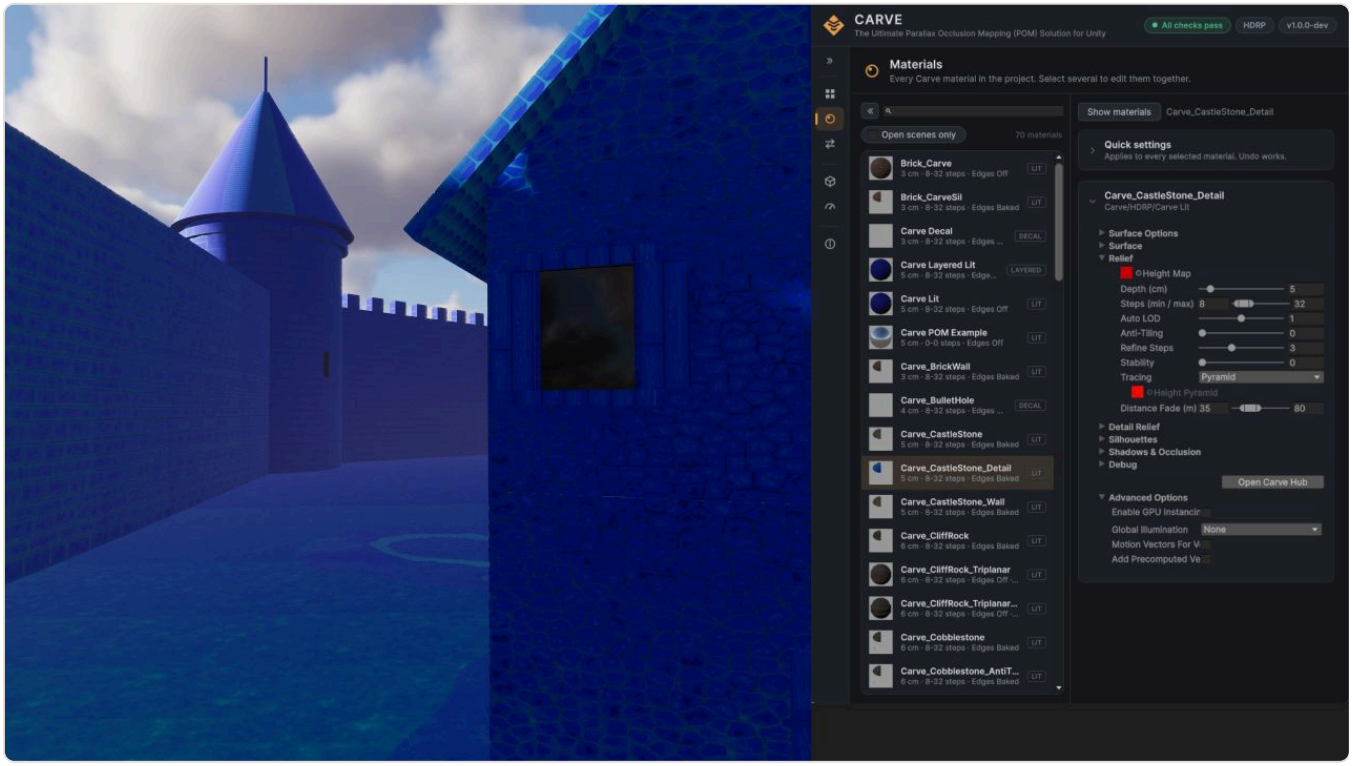
Tools > Carve > Carve Hub is one window for everything Carve. The header pill shows project health; the sidebar badges count materials and conversion candidates.

Page	What it does
Dashboard	Project health with one-click fixes, scene stats, and quick actions.
Materials	Every Carve material in the project. Select several to edit them together: quality presets, depth scaling, the full inspector.
Convert	Lit and LayeredLit to Carve, in place or as copies, and back again.
Meshes	Silhouette data for modular meshes, relief colliders and relief meshes.
Debug	Global debug views, A/B overrides, and the GPU cost meter.
Help	Quick start, the Carve POM node, documentation and support.



The Dashboard lists problems in the open scene with a button for each fix.

18. Debug Views & GPU Cost

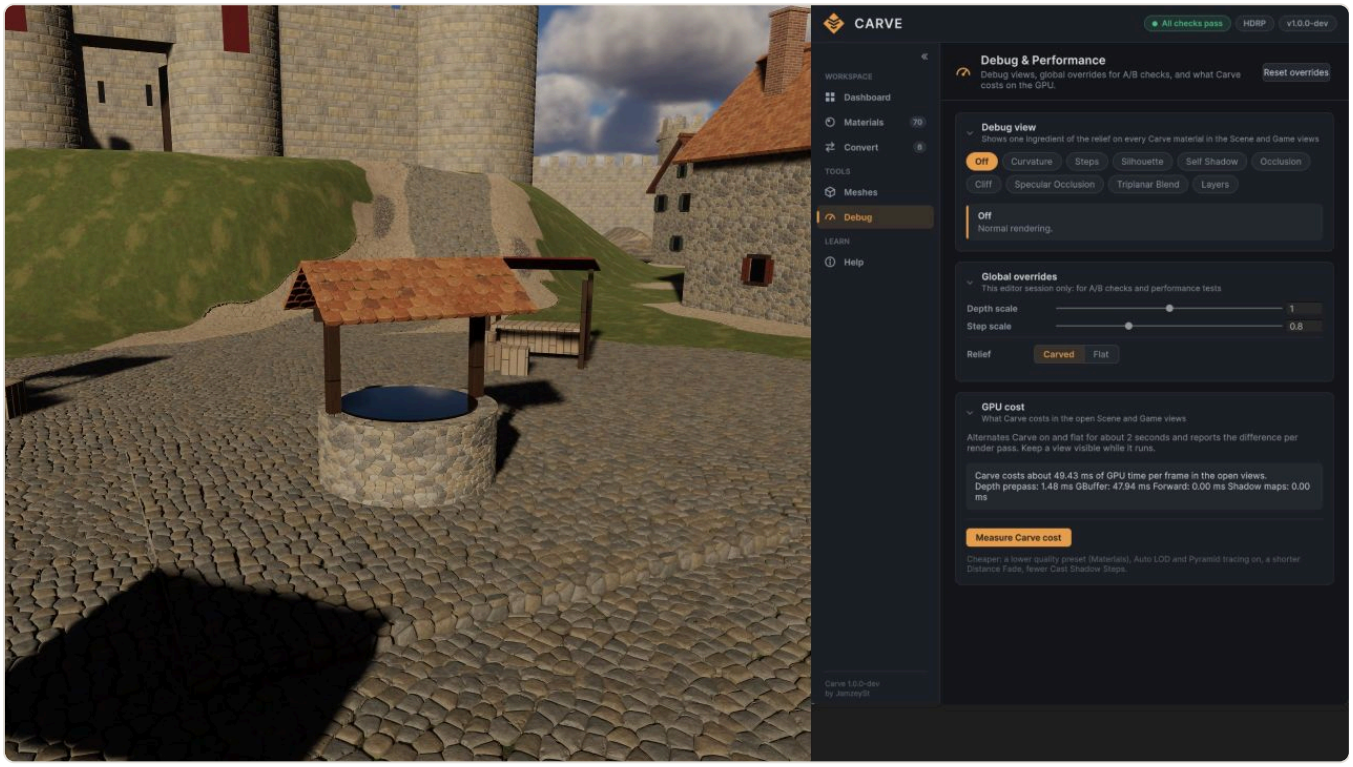


The Steps debug view: blue = few steps, red = many.

View	Shows
Curvature	Red = convex along U, blue = convex along V.
Steps	Ray-march cost per pixel.
Silhouette	Magenta = rays clipped into the outline, gray = the height each ray hit.
Self Shadow	White = lit, black = shadowed by the relief.
Occlusion	White = open, black = occluded.
Cliff	Orange = steep walls where cliff shading takes over.
Specular Occlusion	White = reflections visible, black = blocked.
Triplanar Blend	Red / green / blue = share of the X / Y / Z projection.
Layers	Gray / red / green / blue = layers 0 / 1 / 2 / 3.

Set a view per material (Debug section) or for the whole scene (Hub > Debug). In the demos `V` cycles them.

Cost meter



The cost meter measures Carve on vs flat in the current view.

Measure Carve cost (Hub > Debug > GPU cost) alternates Carve on and flat for about two seconds and reports the median GPU time per render pass (HDRP: depth prepass, GBuffer, forward, shadow maps; URP: depth prepass, depth normals, GBuffer, opaque, main light shadows). It measures your scene, on your GPU, from the current Scene or Game view.

19. Sample Materials



Some of the 13 sample materials.

13 ready-to-use Carve Lit materials for each pipeline (`Materials/HDRP/Samples`, `Materials/URP/Samples`): Cobblestone, Brick Wall, Stone Pavers, Cliff Rock, River Pebbles, Cracked Mud, Wood Planks, Sci-Fi Panels, Roof Tiles, Castle Stone, Fieldstone, Plaster and Turf. Textures are original and procedural, 2048 px and tileable, sized for real-world scale on meter UVs, with height pyramids included.

20. Demo Scenes



Carve_Village: a walled town built with Carve materials.

Scene	Contents
Carve_Showcase	A plaza with one station per feature: silhouettes, real depth, anti-tiling, detail relief, layers, decals, relief collider, triplanar, the Carve POM node and more.
Carve_Village	A walled town: curtain wall with towers and a gatehouse, 34 houses, a market square and an inner castle, on a 240 m four-layer ground, with a Quality Manager.
In URP, open the `_URP` versions in `Demo/Scenes/URP`: the same scenes with URP materials, lighting and post-processing.	
Carve_Minimal	The smallest working setup: a ground, a wall, a column, a block and a rock with sample materials, a sun, a camera and a Volume. Start here to see what a Carve scene needs.

Controls

Key	Action
WASD, Shift	Move, faster. Fly camera: `E` / `Q` up and down, right mouse to look, scroll for speed, `T` tour where available. Walker: right mouse to orbit, scroll to zoom.
C	Carve on / off (flat, normal maps only).
V	Cycle debug views.
L	Sweep the sun.
G	GPU time per pass.
F2 / F3	Quality presets / adaptive (with a Quality Manager in the scene).
F1	Hide or show the help panel.

21. Scripting API

Runtime types are in the `JamzeySt.Carve` namespace (assembly `JamzeySt.Carve.Runtime`); editor tools are in `JamzeySt.Carve.Editor`.

CarveQualityManager

```
var q = CarveQualityManager.Active;          // the scene's manager (null if none)
q.SetPreset(CarveQualityPreset.Medium);      // Low, Medium, High, Ultra, Custom
q.custom.marchSteps = 0.8f;                  // Custom settings (CarveQualitySettings)
q.qualityScale = 0.75f;                      // manual scale (adaptive off)
q.adaptive = true;                           // Play mode: hold targetFrameRate
q.targetFrameRate = 60f;
float scale = q.CurrentScale;                // scale in use
float ms = q.MeasuredFrameMs;                // smoothed frame time (GPU time if q.UsingGpuTime)
```

`CarveQualitySettings.For(preset)` returns a preset's settings. `CarveQualityManager.GpuTimeSource` and `DeltaTimeSource` (static `Func<float>`) replace the GPU timer and the frame clock for tools that drive frames themselves, such as benchmarks or video capture.

CarveGlobals

Scene-wide values read by every Carve material, for settings menus and A/B tests: `DepthScale` (0 = flat), `StepScale`, `DebugView`, feature strengths `SelfShadow`, `Occlusion`, `CliffAndSpecular`, `AntiTiling`, and the quality values the manager sets (`DistanceScale`, `ShadowStepScale`, `ExtrasDistance`, `OcclusionSamples`). `ResetQuality()` returns to every material's own settings.

```
CarveGlobals.DepthScale = 0f;                // Carve off: flat, normal maps only
CarveGlobals.DebugView = CarveDebugView.Steps;
```

CarveSun and CarveLayers

```
CarveSun.Override = myMoonLight;             // light used for self-shadowing
CarveLayers.SetLayerCount(material, 3);      // keeps the compiled layer count in sync
```

Editor tools

Call	What it does
CarveMeshBaker.BakeToAsset(mesh)	Baked copy of a mesh with silhouette data.
CarveReliefBaker.AddColliders(renderers)	Relief colliders; `BakeMeshes(renderers, createObjects)` for relief meshes.
CarveHeightGenerator.GenerateFor(materials)	Generates missing height maps.
CarvePyramidBaker.Assign(materials)	Bakes and assigns height pyramids.
CarveConverter.Plan / Convert	Plan and run a Lit to Carve conversion; `AddRule` adds your own source shaders.
CarveLayerPainter.Begin / Stroke / FillAll / End	Paints layer masks from code.
CarveCostMeter.Start()	Runs the cost meter.

CarveHub.Open(page)	Opens the Hub on a page ("Overview", "Materials", "Convert", "Meshes", "Debug", "Help").
CarveProject.AddCheck(check)	Adds your own health check to the Hub and the Setup Wizard.

22. Performance Guide

- Measure first: the cost meter (Hub > Debug) shows what Carve costs in your view.
- Keep **Tracing = Pyramid** and **Auto LOD** on. They are on in the sample materials.
- Use **Distance Fade** and the Quality Manager's **Relief distance** and **Extras distance** for large scenes.
- Use the cheapest feature that does the job: anti-tiling and detail relief each add samples per step; Carve Layered Lit costs about 2-2.5x Carve Lit at 2 layers, 3x at 3 and 4-5x at 4. Use layers where they are painted, not on every surface.
- Shadow maps: relief finer than a shadow texel costs nothing extra; keep **Cast Shadow Steps** near 0.5.
- Adaptive scales relief work, not resolution, and stops at its Minimum Scale: a view still too heavy there stays above the target.

23. Limitations

- URP Shader Graph has no depth offset: graphs built on the Carve POM node in URP don't write real depth (Carve Lit and Carve Layered Lit in URP do).
- URP uses one occlusion term for ambient light and reflections; Carve's reflection trace is folded into it.
- Self-shadowing comes from the sun (one directional light). Point and spot lights light the relief but don't cast relief self-shadows.
- Edge silhouettes need UV faces (UV Tile or Baked); in triplanar modes only curved outlines are carved.
- Unity's built-in sphere pinches its UVs at the poles: use Triplanar, or a sphere with even UVs.
- Carve Decals give parallax depth but can't write depth.
- Converting LayeredLit matches HDRP's depth approximately (HDRP blends one depth per pixel; Carve keeps each layer's own).
- Anti-tiling can leave a few short steep "walls" at hex seams in grazing close-ups; larger cells make them rarer.
- The Carve POM node covers single-texture relief; triplanar, anti-tiling, detail relief and layers are Carve Lit / Carve Layered Lit features.

24. Troubleshooting

Problem	Fix
The relief looks flat	Check Height Map and Depth (cm). Check the Quality Manager's Relief distance and the material's Distance Fade. In the demos, `C` may have switched Carve off.
Props float above the relief or the relief has no depth	Depth Offset is off, usually left over from Lit. Click **Fix** in the inspector or the Dashboard.
Box edges torn or full of holes	The material uses Baked silhouettes on an unbaked mesh: bake it (Dashboard > Bake now), or use UV Tile.

Shimmering relief	Use TAA on the camera (Dashboard fix). In the Scene view, the fix enables TAA and Always Refresh. Without TAA, raise **Stability**.
No self-shadows	No directional light in the scene, or Self Shadow is 0 on the material or the Quality Manager.
Swirl at the poles of a sphere	UV pinching: switch Mapping to Triplanar Object.
URP: Carve Decals don't show	Add **Renderer Feature > Decal** to the Universal Renderer in use (Dashboard check: Show renderer).
URP: shimmering relief	Set the camera's Anti-aliasing to Temporal Anti-aliasing (Dashboard fix), or raise Stability.
HDRP: the material is drawn twice in the frame debugger	HDRP Alpha Clipping is on: the Dashboard fix turns it off (Carve cuts outlines itself).
A layered material is slower than expected	Its compiled layer count is out of sync: the Dashboard fixes it; from code use `CarveLayers.SetLayerCount`.

25. Package Contents & Generated Files

Folder	Contents
Runtime	Quality Manager, globals, sun, layer helpers.
Editor	Carve Hub, Setup Wizard, inspectors, bakers, converter, height generator, layer painter.
Shaders	Carve Lit, Carve Layered Lit and Carve Decal for HDRP, the shared HLSL, the Carve POM sub-graph and the HDRP example graph.
Materials, Textures	13 sample materials and 4 decals (HDRP) with their textures, shared by both pipelines.
Meshes	Meshes used by the demo scenes.
Demo	HDRP demo scenes, their scripts, materials and settings.
Carve URP Content.unitypackage	URP: Carve Lit, Carve Layered Lit and Carve Decal for URP, the URP example graph, 13 sample materials and 4 decals, and the demo scenes (`Demo/Scenes/URP`) with their materials and settings. Imported by the Setup Wizard.
Documentation	This manual.

Carve writes generated assets to `Assets/Carve Generated/` (Meshes, Pyramids, Relief, Masks), outside the package folder, so updating Carve never overwrites your work. Generated height maps are saved next to their albedo texture.

26. Support

Questions, ideas or a bug? Write to jamzeyst@gmail.com. Please include your Unity version, render pipeline and Carve version; the **Report a bug** button on the Hub's Help page fills them in for you.

Thank you for using Carve.